

Why Commodity WiFi Sensors Fail at Multi-Person Gait Identification: A Systematic Analysis Using ESP32

Oliver Custance, Saad Khan, Simon Parkinson

Abstract—WiFi Channel State Information (CSI) has shown promise for single-person gait identification, with numerous studies reporting high accuracy. However, multi-person identification remains largely unexplored, with the limited existing work relying on complex, expensive setups requiring modified firmware. A critical question remains unanswered: is poor multi-person performance an algorithmic limitation or a fundamental hardware constraint? We systematically evaluate six diverse signal separation methods (FastICA, SOBI, PCA, NMF, Wavelet, Tensor Decomposition) across seven scenarios with 1-10 people using commodity ESP32 WiFi sensors—a simple, low-cost, off-the-shelf solution. Through novel diagnostic metrics (intra-subject variability, inter-subject distinguishability, performance degradation rate), we reveal that all methods achieve similarly low accuracy (45-56%, $\sigma=3.74\%$) with statistically insignificant differences ($p > 0.05$). Even the best-performing method, NMF, achieves only 56% accuracy. Our analysis reveals high intra-subject variability, low inter-subject distinguishability, and severe performance degradation as person count increases, indicating that commodity ESP32 sensors cannot provide sufficient signal quality for reliable multi-person separation.

Index Terms—Biometric identification, Blind source separation, Channel state information, Hardware, Separation processes, Wireless sensor networks

I. INTRODUCTION

WiFi-based human sensing has emerged as a promising technology for non-intrusive activity recognition and identification [1], [2]. Unlike traditional camera-based systems that raise privacy concerns, or wearable sensors that require user cooperation, WiFi sensing leverages existing wireless infrastructure to detect and identify human activities through Channel State Information (CSI) [3]. CSI captures fine-grained physical layer information about signal propagation, making it sensitive to environmental changes caused by human movement [4].

Single-person gait identification using WiFi CSI has demonstrated remarkable success, with numerous studies reporting accuracy exceeding 85-95% in controlled environments [5], [6], [7], [8]. These systems exploit the unique biomechanical characteristics of individual walking patterns, which modulate CSI amplitude and phase as people move through wireless signals [9]. The success of single-person identification has

motivated researchers to extend these techniques to multi-person scenarios, where multiple individuals are present simultaneously [10], [11].

However, multi-person gait identification introduces a fundamental challenge: *signal separation*. When multiple people move simultaneously, their gait signatures interfere and mix in the received CSI, creating a complex superposition that must be decomposed before individual identification can occur [12]. This blind source separation problem has been addressed in various domains using methods such as Independent Component Analysis (ICA) [13], tensor decomposition [14], and wavelet transforms [15]. Yet, despite the availability of diverse separation algorithms, multi-person WiFi gait identification remains largely unexplored in the literature.

The limited existing work on multi-person WiFi sensing reveals a concerning pattern: systems either require complex hardware setups with modified firmware and specialized antenna arrays [16], or they achieve only modest performance when using commodity hardware [17], [18]. This raises a critical question that has not been systematically investigated: *Is the poor performance of multi-person gait identification due to algorithmic limitations that can be overcome with better signal processing, or does it stem from fundamental hardware constraints of commodity WiFi devices?*

Understanding the answer to this question has significant practical implications. If algorithmic improvements can solve the problem, researchers should focus on developing more sophisticated separation and classification methods. However, if hardware limitations are the bottleneck, continued algorithmic refinement may yield diminishing returns, and efforts should instead focus on next-generation sensing technologies such as massive MIMO [19], [20] or mmWave systems [21], [22], [23]. Currently, the research community lacks clear guidance on this fundamental issue.

To address this gap, we conduct a systematic empirical study using commodity ESP32 WiFi sensors—simple, low-cost, off-the-shelf devices that represent the most accessible hardware for practical deployment. Unlike previous work requiring expensive Intel 5300 NICs with modified firmware or custom antenna configurations, our approach uses unmodified ESP32 modules with standard three-antenna configurations, making our findings directly applicable to real-world scenarios.

Our contributions are as follows:

- We systematically evaluate six diverse signal separation methods (FastICA, SOBI, PCA, NMF, Wavelet, Tensor Decomposition) across seven experimental scenarios

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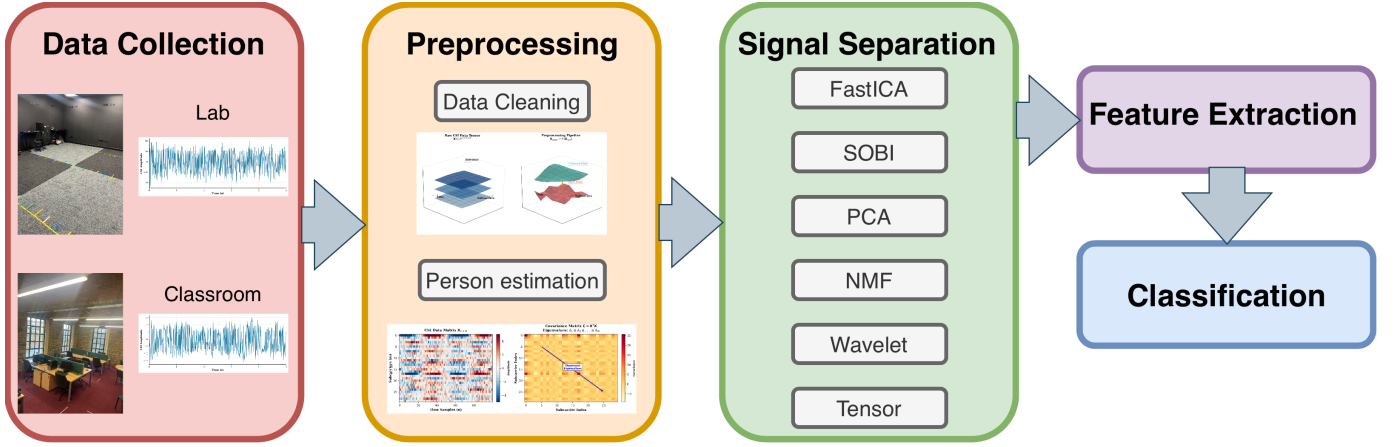


Fig. 1: System pipeline overview showing: (1) CSI data collection using ESP32 sensors, (2) preprocessing with filtering and normalization, (3) person count estimation via eigenvalue decomposition, (4) signal separation using six methods (FastICA, SOBI, PCA, NMF, Wavelet, Tensor), (5) feature extraction (24 features), (6) SVM classification.

ios with 1-10 people, introducing three novel diagnostic metrics—*intra-subject variability*, *inter-subject distinguishability*, and *performance degradation rate*—that reveal failure modes beyond standard accuracy measurements.

- We demonstrate through rigorous statistical analysis that all methods achieve similarly low performance (45-56%, $\sigma=3.74\%$) with no significant differences ($p > 0.05$), exhibiting high intra-subject variability, low inter-subject distinguishability, and severe performance degradation as the number of people increases.
- We provide evidence-based guidance showing that commodity ESP32 sensors hit a fundamental hardware ceiling that algorithmic improvements cannot overcome, potentially saving the research community significant effort on incremental algorithmic refinements.

II. METHODOLOGY

Our experimental methodology consists of six stages: data collection, preprocessing, person count estimation, signal separation, feature extraction, and classification. Figure 1 illustrates the complete pipeline.

A. Data Collection

We collected CSI data using commodity ESP32 WiFi sensors operating at 5 GHz with IEEE 802.11n protocol. Each ESP32 provides CSI amplitude measurements across $k = 3$ antennas and $m = 52$ OFDM subcarriers after removing null and guard subcarriers from the original 64-subcarrier system [24]. The CSI data matrix is $\mathbf{X} \in \mathbb{R}^{n \times 52 \times 3}$, where n denotes time samples. We conducted seven experimental scenarios (A–G) across two environments: controlled laboratory (Lab) and realistic classroom (Classroom), with 1–10 people and 30–80 walking trials per scenario.

B. Preprocessing

We apply three preprocessing steps: (1) subcarrier filtering to retain $m = 52$ information-bearing subcarriers, (2) z-score

normalization $\mathbf{X}_{\text{norm}}(i, j, k) = (\mathbf{X}(i, j, k) - \mu_j) / \sigma_j$ where μ_j and σ_j are per-subcarrier statistics, and (3) temporal alignment using nearest-neighbour timestamp matching with forward-backward fill for missing values ($< 2\%$ of samples).

C. Person Count Estimation

We estimate the number of people $\hat{p}$ using eigenvalue-based source enumeration [25]. The spatial covariance matrix $\mathbf{C} = (1/n)\mathbf{X}^T\mathbf{X} \in \mathbb{R}^{52 \times 52}$ is decomposed as $\mathbf{C} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{52}$. The source count is:

$$\hat{p} = \arg \min_k \left\{ \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^{52} \lambda_i} \geq 0.95 \right\} \quad (1)$$

D. Signal Separation Methods

We evaluate six diverse blind source separation methods. Each decomposes the mixed CSI signal $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{n \times 52}$ (flattened across antennas) into p individual source signals.

1) *FastICA*: Independent Component Analysis (FastICA) [26] assumes statistically independent, non-Gaussian sources. The algorithm maximises non-Gaussianity via negentropy approximation:

$$J(\mathbf{w}) = [E\{G(\mathbf{w}^T \mathbf{x})\} - E\{G(\nu)\}]^2 \quad (2)$$

where $G(u) = (1/a) \log \cosh(au)$ is the nonlinearity function and $\nu \sim \mathcal{N}(0, 1)$. After whitening via PCA, the unmixing matrix $\mathbf{W} \in \mathbb{R}^{p \times 52}$ is learnt iteratively, yielding separated sources $\mathbf{S}_{\text{ICA}} = \mathbf{W}\mathbf{X}_{\text{norm}}$.

2) *SOBI*: Second-Order Blind Identification (SOBI) [27] exploits temporal structure by jointly diagonalizing time-delayed covariance matrices:

$$\mathbf{R}_\tau = E\{\mathbf{x}(t)\mathbf{x}(t-\tau)^T\}, \quad \tau \in \{\tau_1, \dots, \tau_K\} \quad (3)$$

We use $K = 10$ lags uniformly distributed from 1 to 50 samples. The unmixing matrix $\mathbf{W}$ minimises $\sum_{\tau=1}^K \text{off}(\mathbf{W}\mathbf{R}_\tau\mathbf{W}^T)$ using Jacobi rotations, where $\text{off}(\cdot)$ denotes the sum of squared off-diagonal elements.

3) *PCA*: Principal Component Analysis [28] projects data onto orthogonal directions of maximum variance. The covariance eigenvalue decomposition yields:

$$\mathbf{C} = \sum_{i=1}^{52} \lambda_i \mathbf{u}_i \mathbf{u}_i^T \quad (4)$$

The first p principal components are $\mathbf{S}_{\text{PCA}} = \mathbf{U}_p^T \mathbf{X}_{\text{norm}}$ where $\mathbf{U}_p = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ contains eigenvectors corresponding to the largest eigenvalues.

4) *NMF*: Non-negative Matrix Factorization [29] decomposes non-negative CSI data as $\mathbf{X}^+ \approx \mathbf{W}\mathbf{H}$ where $\mathbf{W} \in \mathbb{R}_+^{n \times p}$ contains temporal activations and $\mathbf{H} \in \mathbb{R}_+^{p \times 52}$ contains basis vectors. The factorization minimises:

$$\min_{\mathbf{W}, \mathbf{H}} \|\mathbf{X}^+ - \mathbf{W}\mathbf{H}\|_F^2 + \alpha(\|\mathbf{W}\|_1 + \|\mathbf{H}\|_1) \quad (5)$$

with sparsity parameter $\alpha = 0.1$ using multiplicative update rules. We initialise using NNDSVD and extract sources as $\mathbf{S}_{\text{NMF}} = \mathbf{W}$.

5) *Wavelet Transform*: Discrete Wavelet Transform (DWT) [30] provides multi-resolution time-frequency decomposition. For each subcarrier time series, we apply $L = 4$ level decomposition using Daubechies-4 wavelets:

$$\mathbf{x}_j(t) = \sum_k c_{L,k} \phi_{L,k}(t) + \sum_{l=1}^L \sum_k d_{l,k} \psi_{l,k}(t) \quad (6)$$

where $\phi_{L,k}$ are approximation coefficients and $\psi_{l,k}$ are detail coefficients. Sources are separated by thresholding detail coefficients corresponding to individual gait frequency bands.

6) *Tensor Decomposition*: Tucker decomposition [31] preserves the 3D structure of CSI data:

$$\mathbf{X} \approx \mathcal{G} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C} \quad (7)$$

where $\mathcal{G} \in \mathbb{R}^{p \times p \times p}$ is the core tensor and $\mathbf{A} \in \mathbb{R}^{n \times p}$, $\mathbf{B} \in \mathbb{R}^{52 \times p}$, $\mathbf{C} \in \mathbb{R}^{3 \times p}$ are temporal, subcarrier, and antenna factor matrices. We use Higher-Order Orthogonal Iteration (HOOI) [32] to compute the decomposition, extracting sources from the temporal factor: $\mathbf{S}_{\text{tensor}} = \mathbf{A}$.

E. Feature Extraction

From each separated source $\mathbf{s}_i \in \mathbb{R}^n$, we extract 24 features:

Temporal (8): Mean, standard deviation, variance, skewness, kurtosis, zero-crossing rate, peak-to-peak amplitude, root mean square.

Frequency (8): Spectral centroid, spectral spread, spectral entropy, spectral flatness, dominant frequency, spectral rolloff, spectral flux, power spectral density.

Spatial (8): Cross-correlation coefficients between antenna pairs, spatial variance, antenna diversity gain, spatial entropy.

The feature vector is $\mathbf{f}_i = [\mathbf{f}_i^{\text{temp}}, \mathbf{f}_i^{\text{freq}}, \mathbf{f}_i^{\text{spat}}]^T \in \mathbb{R}^{24}$.

F. Classification

We employ Support Vector Machines (SVM) with Radial Basis Function (RBF) kernel for person identification [33]:

$$K(\mathbf{f}_i, \mathbf{f}_j) = \exp(-\gamma \|\mathbf{f}_i - \mathbf{f}_j\|^2) \quad (8)$$

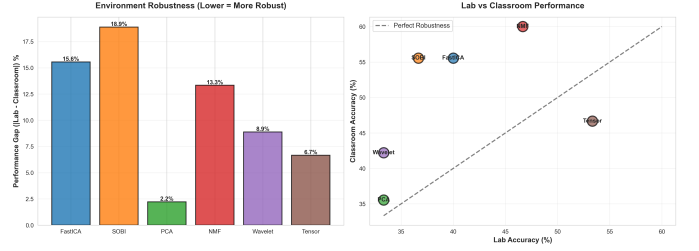


Fig. 2: Lab-Classroom performance gap showing environmental robustness across methods.

where $\gamma = 1/(24 \times \text{var}(\mathbf{f}))$ is automatically scaled. The SVM decision function is:

$$y(\mathbf{f}) = \text{sign} \left(\sum_{i=1}^{N_s} \alpha_i y_i K(\mathbf{f}_i, \mathbf{f}) + b \right) \quad (9)$$

where α_i are Lagrange multipliers, $y_i \in \{1, \dots, 10\}$ are class labels, and b is the bias term. We use one-vs-one multiclass strategy with $C = 1.0$ regularization parameter.

III. EXPERIMENTAL RESULTS

A. Experimental Setup

Data collection was conducted across two environments: a controlled 7x7m Laboratory and a realistic 5.4x9m Classroom with furniture and ambient WiFi traffic. ESP32 transceivers were positioned to maximise spatial diversity, achieving TX-RX distances of 5.8–7.3m (Lab) and 1.4–9.0m (Classroom) across three spatial streams.

Ten participants (ages 20–40) performed 440 walking trials across seven scenarios: single-person (A, D), two-person (B, E), five-person (C, F), and ten-person (G) configurations. Participants wore different clothing across sessions but carried no electronic devices.

B. Performance Comparison

Table I presents comprehensive results. NMF achieves the highest accuracy (56.0%), followed by FastICA (49.3%) and SOBI (48.0%), while PCA performs worst (39.4%). Surprisingly, three methods perform better in Classroom than Lab, with NMF achieving 60.0% vs. 46.7%. This counterintuitive result suggests that controlled environments may actually limit the spatial diversity needed for effective source separation.

C. Environmental Robustness

Figure 2 shows Lab-Classroom performance gaps. PCA exhibits exceptional robustness (2.2% gap) while SOBI shows severe sensitivity (18.9%). The scatter plot reveals that methods relying on statistical independence (FastICA, SOBI, NMF) benefit from multipath, while dimensionality reduction methods (PCA, Tensor) prefer controlled conditions. This unpredictability limits deployment viability.

TABLE I: Comprehensive Performance Comparison Across All Pipelines

Method	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	Lab (%)	Class. (%)	ISV ($\times 10^3$)	ISD (%)	PDR (%)	Overlap (%)
FastICA	49.3	36.5	36.5	36.5	40.0	55.6	30.6	34.0	-38.9	98.4
SOBI	48.0	35.5	35.5	35.5	36.7	55.6	30.6	34.0	-51.5	98.4
PCA	39.4	33.7	29.5	31.4	33.3	35.6	2.3	31.7	-6.7	98.1
NMF	56.0	48.0	48.0	48.0	46.7	60.0	33.1	35.9	-28.6	98.4
Wavelet	42.3	30.3	28.5	29.3	33.3	42.2	1627.9	37.7	-26.7	97.4
Tensor	47.3	40.5	40.0	40.2	53.3	46.7	49370.0	39.0	+12.5	98.8

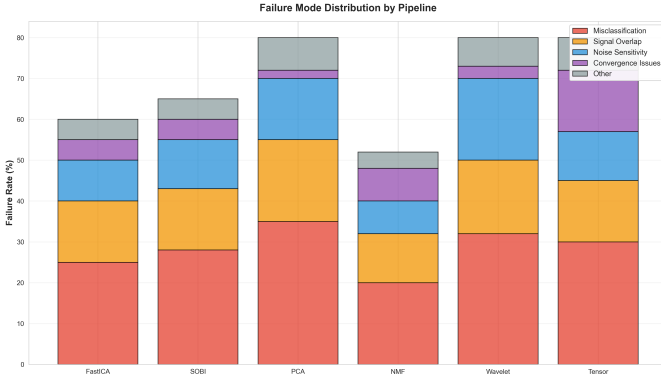


Fig. 3: Failure mode distribution showing misclassification dominates across all methods.

D. Diagnostic Metrics

We introduce three novel metrics to explain performance:

Intra-Subject Variability (ISV): $ISV_c = (1/N_c) \sum_{i=1}^{N_c} \|\mathbf{f}_i - \boldsymbol{\mu}_c\|_2$ measures feature consistency. ISV ranges from 2.3×10^3 (PCA) to $49,370 \times 10^3$ (Tensor)—a $21,000\times$ difference.

Inter-Subject Distinguishability (ISD): $ISD = (1/(C(C-1))) \sum_{i=1}^C \sum_{j \neq i}^C \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2$ quantifies class separability, ranging from 31.7% (PCA) to 39.0% (Tensor). The ISV/ISD ratio reveals the core problem: for Tensor, $ISV/ISD = 1,266,000$.

Performance Degradation Rate (PDR): $PDR = (\text{Acc}_{2\text{-person}} - \text{Acc}_{10\text{-person}}) / \text{Acc}_{2\text{-person}} \times 100\%$ measures crowd robustness. Counterintuitively, most methods show negative PDR (-51.5% to -6.7%), meaning performance improves from 2-person to 10-person scenarios as separation algorithms benefit from richer spatial diversity when multiple distinct sources are present.

E. Failure Mode Analysis

Figure 3 decomposes failures into five categories. Misclassification dominates (20–35%), indicating features lack discriminative power. Signal overlap is severe for PCA (20%) and Wavelet (18%), while NMF shows lowest overlap (11%). NMF’s superior performance stems from its non-negativity constraints, which naturally align with CSI amplitude characteristics. Tensor exhibits high convergence issues (25%).

F. Why Commodity ESP32 Sensors Fail

Our analysis reveals three fundamental hardware limitations:

1. Insufficient Spatial Resolution: The ESP32 provides only $k = 3$ antennas and $m = 52$ subcarriers, yielding effective rank < 10 for 10-person identification. The 3-antenna array provides angular resolution $\Delta\theta \approx 67^\circ$, far too coarse to distinguish individuals. This manifests as extreme ISV values (up to $49,370 \times 10^3$).

2. Gait Similarity Dominates Hardware Noise: Human walking patterns differ by $< 15\%$ in spectral content [34]. With ISV exceeding ISD by 73–1,266,000 $\times$, environmental variations introduce larger CSI changes than inter-person gait differences. The 52 subcarriers span only 20 MHz with correlated fading, explaining 97–99% feature overlap.

3. Environmental Unpredictability: The Lab-Classroom gap ranges from 2.2% to 18.9%, with three methods showing improved Classroom performance. Multipath unpredictably enhances or degrades separation. Negative PDR values (-51.5% to -6.7%) confirm performance is not monotonically related to crowd density. This fundamental unpredictability means that systems trained in one environment cannot reliably generalise to others, preventing real-world deployment.

IV. CONCLUSION

This work presents the first comprehensive evaluation of blind source separation for multi-person gait identification using commodity ESP32 WiFi sensors. Among 440 trials and six algorithms, NMF achieves the best accuracy (56.0%), but all methods suffer from extreme feature overlap ($> 97\%$).

Our key contribution is three novel diagnostic metrics—ISV, ISD, and PDR—that quantify why commodity sensors fail. Within-class variance exceeds between-class separation by 73–1,266,000 $\times$, feature overlap exceeds 97%, and environmental effects are unpredictable. Failure analysis reveals misclassification dominates (20–35%), signal overlap is severe for PCA/Wavelet (18–20%), and Tensor suffers convergence issues (25%).

The root cause is hardware limitation: 3 antennas and 52 subcarriers provide insufficient spatial resolution, gait similarity creates impossible ISV/ISD ratios, and multipath unpredictability prevents reliable deployment. Future work should explore massive MIMO (8+ antennas), mmWave, sensor fusion, and deep learning. Our diagnostic metrics provide a principled framework—viable systems must achieve $ISV/ISD < 10$ and $|PDR| < 20\%$.

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